CHM 120

Chemical Reactions

Final Report

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Test Your Knowledge

Categorize each change as physical change or chemical change.

Physical Change	**Chemical Change**
1	2
Rubbing alcohol evaporates	A screwdriver left outside in the rain rusts
Water boils	Silver tarnishes upon exposure to the air
Sugar dissolves in water	Barium hydroxide mixed with ammonium chloride becomes colder
	A butane lighter produces a flame

Identify the coefficients that balance each chemical equation.

__$C_3H_8(g)$ + __$O_2(g)$ __$CO_2(g)$ + __$H_2O(l)$

1 1, 5, 3, 4

__$Zn(s)$ + __$HCl(aq)$ __$ZnCl_2(aq)$ + __$H_2(g)$

2 1, 2, 1, 1

__$NaOH(aq)$ + __$H_2SO_4(aq)$ __$Na_2SO_4(aq)$ + __$H_2O(l)$

3 2, 1, 1, 2

__$Na_2SO_4(aq)$ + __$BaCl_2(aq)$ __$BaSO_4(s)$ + __$NaCl(aq)$

4 1, 1, 1, 2

Classify each chemical reaction.

1 Decomposition $2NaCl \rightarrow 2Na + Cl_2$

2 Combination $2Na + Cl_2 \rightarrow 2NaCl$

3 Single displacement $Li + NaCl \rightarrow LiCl + Na$

4 Double displacement $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

Arrange the metals from least active to most active.

Li - Lithium
K – Potassium
Ba – Barium
Sr – Strontium
Ca – Calcium
Na – Sodium
Mg – Magnesium
Al – Aluminum
Mn – Manganese
Zn – Zinc
Cr – Chromium
Fe – Iron
Cd – Cadmium
Co – Cobalt
Ni – Nickel
Sn – Tin
Pb – Lead
H_2 – Hydrogen
Cu – Copper
Ag – Silver
Hg – Mercury
Au - Gold

- Lead
- Zinc
- Magnesium
- Calcium
- Potassium

Exploration

A chemical reaction occurs when the atoms of substances interact with one another, rearranging to form new chemical substances.

- (•) True
- ( ) False

In the chemical reaction 8Fe + S_8 yields 8FeS, Fe is a _____, while FeS is a _____.

- (selected) reactant; product
- product; reactant
- Both Fe and FeS are reactants.
- Both Fe and FeS are products.

Chemical changes include bread molding, putting hydrogen peroxide on a cut, and boiling water.

- True
- (selected) False

How many NaCl are required to balance the equation?

$$NaCl + CuSO_4 \rightarrow Na_2SO_4 + CuCl_2$$

- 1
- (selected) 2
- 3
- 4

Classify each reaction as decomposition, combination, single displacement, double displacement, or acid-base neutralization.

a. $Zn(s) + CuSO_4(aq) \rightarrow Cu(s) + ZnSO_4(aq)$

b. $HNO_3(aq) + NaOH(aq) \rightarrow NaNO_3(aq) + H_2O(l)$

c. $Pb(NO_3)_2(aq) + 2KCl(aq) \rightarrow 2KNO_3(aq) + PbCl_2(s)$

d. $2H_2O_2(aq) \rightarrow O_2(g) + 2H_2O(l)$

- a. double displacement; b. decomposition; c. single displacement; d. acid-base neutralization
- a. acid-base neutralization; b. decomposition; c. double displacement; d. single displacement
- a. acid-base neutralization; b. single displacement; c. double displacement; d. decomposition
- a. single displacement; b. acid-base neutralization; c. double displacement; d. decomposition

List metals from least active to most active.

- Ag < Al < Cu < K < Mg < Zn
- Cu < K < Ag < Al < Mg < Zn
- Ag < Cu < Zn < Al < Mg < K
- Zn < Ag < Cu < K < Al < Mg

_____ describe(s) why an event occurred.

- Observations and conclusions
- An observation
- A conclusion

Indicators of chemical reactions include changes in color and the formation of a precipitate or gas.

- True
- False

Exercise 1

Which of the reactants in Data Table 1 showed evidence of a chemical reaction with the aluminum foil? Use your observations to explain your answer.

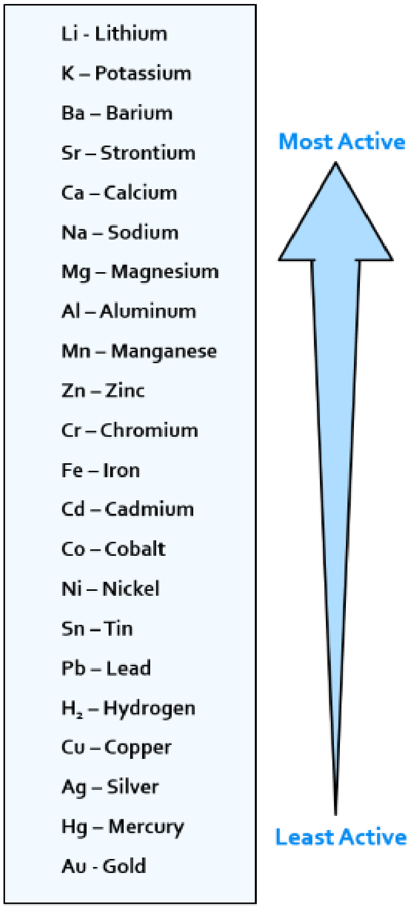
B *I* U 43 Word(s)

$AgNo_3$

$CuSO_4$

These two seemed to be the ones that left a pattern of their own, while the other two just seemed to remain clear. Even after the foil was rinsed, there was a pattern left behind for the two reactants listed above.

List the metals found in each of the four reactants in Data Table 1. Indicate the metal or metals that are more active than aluminum. (See the activity series below.)



B I U 25 Word(s)

The metals are Ag (silver), Cu (copper), and Ca (calcium), and Pb (lead). The one's that are move active than aluminum are sodium and calcium.

Use the activity series to determine whether aluminum would replace the metals in the chemical compounds listed in Data Table 1. Does the activity series' ability to predict whether or not a reaction will occur agree with your observations?

B I U 26 Word(s)

I don't believe that it would be able to replace the metals. The metals are simply all over the chart that are in Data Table 1.

What are the potential sources of error that might cause disagreement between the activity series' prediction of reactions and your observations of reactions?

B I U 32 Word(s)

There could have been multiple variables that could have caused error when it came to the observations. Not enough time to let things react, not enough light, maybe temperature, clean environment, etc...

Write a balanced equation for the reaction between silver nitrate and aluminum. What type of reaction is this?

B I U 4 Word(s)

$AgNO_3$ + Al = Ag + $Al(NO_3)_3$

Data Table 1: Observations of Chemical Reactions with Aluminium Foil

	$AgNO_3$
Start Time	4:38pm
Initial Appearance of Chemical	clear and kind of smudgy
Initial Appearance of Foil	somewhat glitter-like maybe from salt mix
Observations after 5 Minutes	more blackish smudge started to gather
Observations after 3 Hours	very speckled with smudge of a blackish color
Appearance of Foil after Rinsing	no longer black nor smudgy; left a sort of salty look behind

	$CuSO_4$
Start Time	4:38pm
Initial Appearance of Chemical	clearish-blue with bubbles
Initial Appearance of Foil	somewhat glitter-like maybe from salt mix
Observations after 5 Minutes	brownish color started to appear
Observations after 3 Hours	the brownish color is now rusty looking with some spots appearing larger than others
Appearance of Foil after Rinsing	left behind speckles of black throughout the area that the drops covered;

	$CaCl_2$
Start Time	4:38pm
Initial Appearance of Chemical	just clear and very stable
Initial Appearance of Foil	somewhat glitter-like maybe from salt mix
Observations after 5 Minutes	

	clear and somewhat thick seeming
Observations after 3 Hours	still just looks clear; no change really
Appearance of Foil after Rinsing	there is nothing really to see; everything is gone

	$Pb(NO_3)_2$
Start Time	4:38pm
Initial Appearance of Chemical	also just clear
Initial Appearance of Foil	somewhat glitter-like maybe from salt mix
Observations after 5 Minutes	also clear, doesn't seem to have change really
Observations after 3 Hours	still clear as well; kind of cloudy, but really no change
Appearance of Foil after Rinsing	there is nothing really to see; everything is gone

Exercise 2

Reviewing Data Table 2, what were the visual cues that chemical reactions occurred?

B *I* U 6 Word(s)

Colors mainly, a few textures too.

Define a single displacement reaction and a double displacement reaction. Describe the similarities and differences between a single displacement reaction and a double displacement reaction.

B I U 34 Word(s)

Basically, for a single displacement reaction, one element is exchanged with another one, inside of whatever compound.
Now, a double displacement reaction deals with two elements being exchangee with two others in whatever compound.

Data Table 2 contains nine double displacement reactions. Write a balanced chemical equation for five of those reactions.

B I U 0 Word(s)

Which reactions produced a precipitate? Describe the observations when answering this question.

B I U 10 Word(s)

Silver Nitrate and Aqueous Ammonia
Lead Nitrate and Calcium Chloride

Which reactions produced a color change? Describe the change in color for each of these reactions. (For example, the reactants were clear and colorless and the product was a bright pink solution.)

B *I* U 79 Word(s)

IKI indicator with Starch turned to black.
KI Potassium Iodide and Lead(II) Nitrate turned green and brown.
Sodium Hydroxide with Phenolphthalein turned pink-purple.
Sodium Hydroxide with Silver(II) Nitrate turned muddy brown
Silver Nitrate and Aqueous Ammonia turned white and milky.
Aqueous Ammonia and Copper(II) Sulfate turned lighter blue.
Lead Nitrate and Calcium Chloride turned milky white.
Copper(II) Sulfate and Sodium Bicarbonate turned light blue.
Blue Dye #1 and Hydrochloric Acid turned dark green.
Sodium Carbonate with Phenolphthalein turned purple

Data Table 2: Reaction Observations

Well	Chemical #1 (4 drops)
A1	$NaHCO_3$ Sodium Bicarbonate
A2	IKI indicator
A3	KI Potassium Iodide
A4	NaOH Sodium Hydroxide
A5	HCl Hydrochloric Acid
A6	NaOH Sodium Hydroxide
B1	$AgNO_3$ Silver Nitrate
B2	NH_4OH Aqueous Ammonia
B3	$Pb(NO_3)_2$ Lead Nitrate
B4	$NaHSO_4$ Sodium Bisulfate
B5	$NaHSO_4$ Sodium Bisulfate
B6	$CuSO_4$ Copper(II) Sulfate
C1	Blue Dye #1
C2	Na_2CO_3 Sodium Carbonate

Well	Chemical #1 Appearance
A1	clear; very thin liquid
A2	orange-brown; seems sticky, adhering to side of well
A3	orange-brown; seems sticky, adhering to side of well
A4	clear
A5	clear; somewhat cloudy
A6	clear
B1	clear
B2	clear
B3	clear
B4	clear
B5	

	clear
B6	clear-blue
C1	darker blue
C2	clear

Well	Chemical #2 (4 drops)
A1	HCl Hydrochloric Acid
A2	Starch
A3	$Pb(NO_3)_2$ Lead(II) Nitrate
A4	$C_{20}H_{14}O_4$ Phenolphthalein
A5	$C_{20}H_{14}O_4$ Phenolphthalein
A6	$AgNO_3$ Silver(II) Nitrate
B1	NH_4OH Aqueous Ammonia
B2	$CuSO_4$ Copper(II) Sulfate
B3	$CaCl_2$ Calcium Chloride
B4	$CaCl_2$ Calcium Chloride
B5	Na_2CO_3 Sodium Carbonate
B6	$NaHCO_3$ Sodium Bicarbonate
C1	HCl Hydrochloric Acid
C2	$C_{20}H_{14}O_4$ Phenolphthalein

Well	Chemical #2 Appearance
A1	clear; somewhat cloudy
A2	clear, a bit foggy
A3	clear
A4	clear
A5	clear
A6	clear
B1	clear
B2	

	clear-blue
B3	clear
B4	clear
B5	clear
B6	clear
C1	clear
C2	clear

Well	Observations
A1	They started to fizz and bubble once mixed.
A2	immediately turned black; like black, dirty paint water
A3	there is a farily gross green and brown mixture; also it seems to be sparkling and has an area that looks like just lead
A4	turned a very pretty pink-purple;
A5	it just seemed to get foggy; a bit fizzy
A6	brown and very muddy looking
B1	Observation 1: milky looking; thick + Absorb in paper towel and expose to sunlight Observation 2: somewhat yellow looking
B2	cloudy light blue; sticky
B3	milky but thin
B4	it still looks pretty clear

B5	bubbles for sure; I swear I smelt something, could be wrong
B6	very bowlingball looking, like light blue and clear swirls
C1	turned a darker green but also still clear
C2	turned purple quickly and I swear I smelt hand-sanitizer

Well	Chemical Change (Yes/No)
A1	Yes, I believe so.
A2	Yes.
A3	Yes.
A4	Yes
A5	No.
A6	Yes.
B1	Yes.
B2	No.
B3	Yes.
B4	No.
B5	Yes.
B6	Yes.
C1	Yes.
C2	Yes.

Competency Review

During a chemical reaction, atoms of chemicals _____.

- interact with one another
- rearrange to form new substances
- both interact with one another and rearrange to form new substances

During a chemical reaction, _____ will yield _____.

- reactants; products
- products; reactants

_____ is NOT necessarily the result of a chemical reaction.

- A hard-boiled egg
- A fire
- Steam
- A rusted shovel

A _____ is a change in the form or state of a substance.

- chemical change
- physical change
- reaction

Choose the correct coefficients for the following reaction: _____ H_2 + _____ N_2 *yields* $2NH_3$.

- 2; 3
- 3; 2
- 3; 1
- 1; 2

_____ is a combination reaction.

- 2NaCl *yields* 2Na + Cl_2
- Li + NaCl *yields* LiCl + Na
- 2Na + Cl_2 *yields* 2NaCl

An activity series consists of _____ ordered by _____.

- compounds; activity
- elements; size
- elements; activity
- compounds; size

Observations are a descriptions of what happened and why.

- True
- False

A change in color can indicate a chemical reaction.

- True
- False

If a clear liquid is mixed with a green liquid and no noticeable changes are observed, the most appropriate observation would be _____.

- no reaction occurred
- the result was a green liquid and no observable changes occurred
- the clear reactant turned green

If two liquids are combined and a precipitate forms, it is likely that a chemical reaction has occurred.

- True
- False

When H_2SO_4 is mixed with NaOH, an acid-base neutralization reaction proceeds, and _____ is one of the products. Hint: Write the balanced equation before attempting to select an answer.

- H_3O
- Na_2SO_4
- Na_2S
- $NaSO_3$

List the coefficients for the reaction: $C_{10}H_8$ + O_2 yields CO_2 + H_2O.

- 1, 8, 6, 2
- 1, 12, 10, 4
- 1, 4, 6, 8
- 1, 10, 12, 8

Aluminum replacing a less active metal in a compound is classified as a _____ reaction.

- decomposition
- single displacement
- combination
- double displacement
- neutraliztion

Chemical changes always occur very quickly, like explosions.

- True
- False

Extension Questions

Give three examples each of physical and chemical changes you observed in the last 24 hours. Explain why you classified each example as physical or chemical.

Three physical changes: water turning into ice; water boiling for noodles; my juice drink unfreezing after being left in the freezer for too long

Three chemical changes: wood burning; toast being made; salt and water mixture